2 The Tasks of the Test Analyst in Risk-Based Testing – 90 minutes

Keywords

impact analysis, product risk, regression testing, risk analysis, risk assessment, risk control, risk identification, risk mitigation, risk monitoring, risk-based testing

Learning Objectives for Chapter 2:

2.1 Risk Analysis

TA-2.1.1 (K2) Summarize the test analyst's contribution to product risk analysis

2.2 Risk Control

TA-2.2.1 (K4) Analyze the impact of changes to determine the scope of regression testing

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Introduction to the Tasks of the Test Analyst in Risk-Based Testing

Risk-based testing is a test approach that prioritizes test efforts based on the risk levels of the test items. The test manager determines this approach. The TA plays an active role in implementing it. Risk-based testing is covered in more detail in (ISTQB-TM, v3.0).

Risk-based testing involves risk analysis and risk control (ISTQB-CTFL, v4.0.1, Section 5.2).

In addition, since risks and risk levels are not static and change over time, risk-based testing involves regular risk monitoring. In an iterative lifecycle, this is performed at a frequency determined by the team (likely to be once per iteration). In other lifecycles, its frequency is set by the person responsible for product risk management, often the test manager. The TA contributes by updating the risk register based on the changes made and adjusting the risk mitigation actions mentioned above.

2.1 Risk Analysis

Risk analysis involves risk identification and risk assessment. This syllabus focuses on how the TA should participate in risk analysis activities to ensure risk-based testing is implemented correctly.

The TA often possesses unique knowledge of the system, as well as experience and intuitive knowledge of what usually goes wrong, what impact it has, and how testing can mitigate the risk. This makes the TA a valuable stakeholder for product risk analysis.

In **risk identification**, the TA contributes with their own experience and knowledge to retrospectives, risk workshops, brainstorming, and creating checklists. The TA can also conduct interviews with stakeholders to better understand what they consider to be the most significant risks from their perspective.

During **risk assessment**, the TA, together with other stakeholders, contributes to the determination of the risk level by estimating several factors such as:

- frequency of use and criticality of the affected features
- criticality of the affected business objectives
- financial, environmental, and reputational damage
- quality of the test basis
- legal or safety needs

The TA also helps categorize product risks by the quality characteristics impacted, for example, using the product quality model of *ISO/IEC 25010* (2023). The risk level is often not uniformly distributed across the test object. In such cases, the TA should break down the test object into test items (e.g., components, interfaces, and features) and assess a given risk for each test item separately.

Finally, as part of the product risk assessment, the TA proposes suitable test activities to mitigate each identified product risk. These activities may include static testing and dynamic testing. Depending on factors such as the risk level, the associated test item type, and the quality characteristic affected, the TA indicates the required test levels, test types, test techniques, levels of independence of testing, and test thoroughness. In the spirit of shift left, the TA indicates which test activities can mitigate the risk earliest to minimize the testing effort.

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